

Parker Seegmiller

Hanover, NH, USA · (385) 219-8093 · m.parker.seegmiller@gmail.com · pkseeg.com

Selected Publications

1. **Seegmiller, P.** & Preum, S. (2026). “Measuring distribution shift in user prompts and its effects on LLM performance” *ACL 2026*.
2. **Seegmiller, P.**, Gatto, J., Greer, S., Isingizwe, G., Ray, R., Burdick, T., & Preum, S. (2026). “How much would a clinician edit this draft? Evaluating LLM alignment for patient message response drafting” *ACL 2026*.
3. **Seegmiller, P.**, Mehta, K., Saha, S., Tao, C., Oraby, S., Gupta, A., Chung, T., Bansal, M., & Peng, N. (2025). “FLAMES: Improving LLM math reasoning via a fine-grained analysis of the data synthesis pipeline” *Findings of EMNLP 2025*.
4. **Seegmiller, P.** & Preum, S. (2023). “Statistical depth for ranking and characterizing transformer-based text embeddings” *EMNLP 2023*.
5. **Seegmiller, P.**, Gatto, J., Johnston, G., Basak, M., & Preum, S. (2023). “HealthE: Recognizing health advice & entities in online health communities” *ICWSM 2023*.

Additional Publications

6. Gatto, J., **Seegmiller, P.**, Burdick, T. E., Khayal, I. S., DeLozier, S., & Preum, S. M. (2025). “Follow-up question generation for enhanced patient-provider conversations” *ACL 2025*.
7. Gatto, J., Sharif, O., **Seegmiller, P.**, & Preum, S. M. (2025). “Document-level event-argument data augmentation for challenging role types” *ACL 2025*.
8. **Seegmiller, P.**, Gatto, J., Sharif, O., Basak, M., & Preum, S. M. (2024). “Do LLMs find human answers to fact-driven questions perplexing? A case study on Reddit” *Workshop at ICWSM 2024*.
9. Gatto, J., **Seegmiller, P.**, Burdick, T. E., & Preum, S. M. (2024). “In-context learning for preserving patient privacy: A framework for synthesizing realistic patient portal messages” *Machine Learning for Health (ML4H 2024)*.
10. Amazon AGI Foundations. (2024). “The Amazon Nova family of models: Technical report and model card” *Amazon (contributor)*.
11. Gatto, J., Sharif, O., **Seegmiller, P.**, Bohlman, P., & Preum, S. M. (2023). “Text encoders lack knowledge: Leveraging generative LLMs for domain-specific semantic textual similarity” *Workshop at EMNLP 2023*.
12. **Seegmiller, P.**, Gatto, J., Basak, M., Cook, D., Ghasemzadeh, H., Stankovic, J., & Preum, S. (2023). “The scope of in-context learning for the extraction of medical temporal constraints” *HealthNLP Workshop, ICHI 2023*.
13. Gatto, J., **Seegmiller, P.**, Johnston, G., & Preum, S. (2022). “Identifying the perceived severity of patient-generated telemedicine queries regarding COVID” *Journal of Medical Internet Research, 2022*.

Experience

Research Scientist Intern

June 2026 – August 2026

Collinear AI, Sunnyvale, CA

- Agentic large language models

Applied Science Intern

August 2024 – December 2024

Amazon — Foundations of AGI Team, Sunnyvale, CA

- Developed data synthesis pipeline for LLM mathematical reasoning; contributor to Amazon Nova, a deployed family of commercial foundation models (see: FLAMES, EMNLP 2025)

Teaching Assistant*September 2021 – Present**Dartmouth College, Department of Computer Science, Hanover, NH*

- Machine Learning, Discrete Mathematics, Android Programming

Data Science Intern*June 2020 – August 2020**Aetna, a CVS Health Company (Payment Integrity Team), Remote*

- Built ML model predicting healthcare provider abusive upcoding on inpatient DRG claims, projected to save up to \$1,000,000/month via audit recommendations

Research Assistant*September 2018 – December 2020**Brigham Young University, Provo, UT*

- ML and statistical research across CS, Marketing, and Statistics departments (fake news classification, sports analytics, behavioral science modeling)

CS Instructor*July 2019 – August 2021**Juni Learning, Inc., Remote*

- Mentored 20+ students in Python and Java; prepared students for USA Computing Olympiad

Teaching Assistant*January 2019 – April 2021**Brigham Young University, Departments of CS & Statistics, Provo, UT*

- Introductory data science, statistics, and programming courses

Education

PhD, Computer Science*September 2021 – November 2026**Dartmouth College, Hanover, NH*

- Research focus: reliability and robustness of LLMs in high-stakes deployment settings, with applications to responsible AI in patient-provider portal communication
- Advisor: Prof. Sarah Masud Preum
- Awards: Guarini PhD Innovation Fellow; Guarini School Graduate Student Poster Session 2023 Winner

BS, Statistics: Data Science*August 2017 – April 2021**Brigham Young University, Provo, UT*

- Minors: Computer Science, Mathematics

Skills

Programming: Python, Java, C++, SQL, R**Frameworks & Libraries:** PyTorch, HuggingFace Transformers, vLLM, sentence-transformers, scikit-learn, pandas, NumPy, SciPy**LLM & NLP:** Fine-tuning (SFT), preference learning (RLHF/DPO), retrieval-augmented generation (RAG), agentic AI systems, AI agent reliability, LLM reasoning, chain-of-thought prompting, instruction tuning, prompt engineering, text embeddings, synthetic data generation, data mixing & optimization, distributional robustness, responsible AI, AI safety, model governance, evaluation methodology, clinical NLP, benchmark development**Infrastructure:** Git, Linux/Bash, SLURM, Weights & Biases (W&B), MLflow, Jupyter**Human Languages:** English (Native), Lithuanian (Advanced)**Service**

Reviewer: NeurIPS (2025–2026), ACL (2026), EMNLP (2023–2025)